12/01/2026

EE–556 NUCLEAR REACTOR CONTROL

Lecture–2: Reactor Physics Preliminaries

A P Tiwari, FNAE

Visiting Professor

School of Computing and Electrical Engineering

Indian Institute of Technology Mandi

Kamand (HP), India

$$^{1}_{0}n + ^{235}_{92}U \longrightarrow \text{Fission} + \nu \text{ neutrons} + 192\,\text{MeV energy}$$

2 or 3

Notations and Terminologies

- Z = Atomic Number = Number of protons in a nucleus
- A = Mass number = Number of nucleons (protons + neutrons) in a nucleus
- N = Number of neutrons in a nucleus; $N = A - Z$
- ${}^{A}_{Z}\mathrm{X}$ = a specific nucleus with the chemical symbol X. Also denoted as (A, Z)
- Nuclides: Various species of nuclei
- Isotopes: nuclei having the same Z but different A
- ${}^{A}_{Z}\mathrm{X}^{*}$ = nucleus in excited state
- ${}^{A}_{Z}{}^{m}\mathrm{X}$ = isomer, i.e., long lived excited state of nucleus X

n

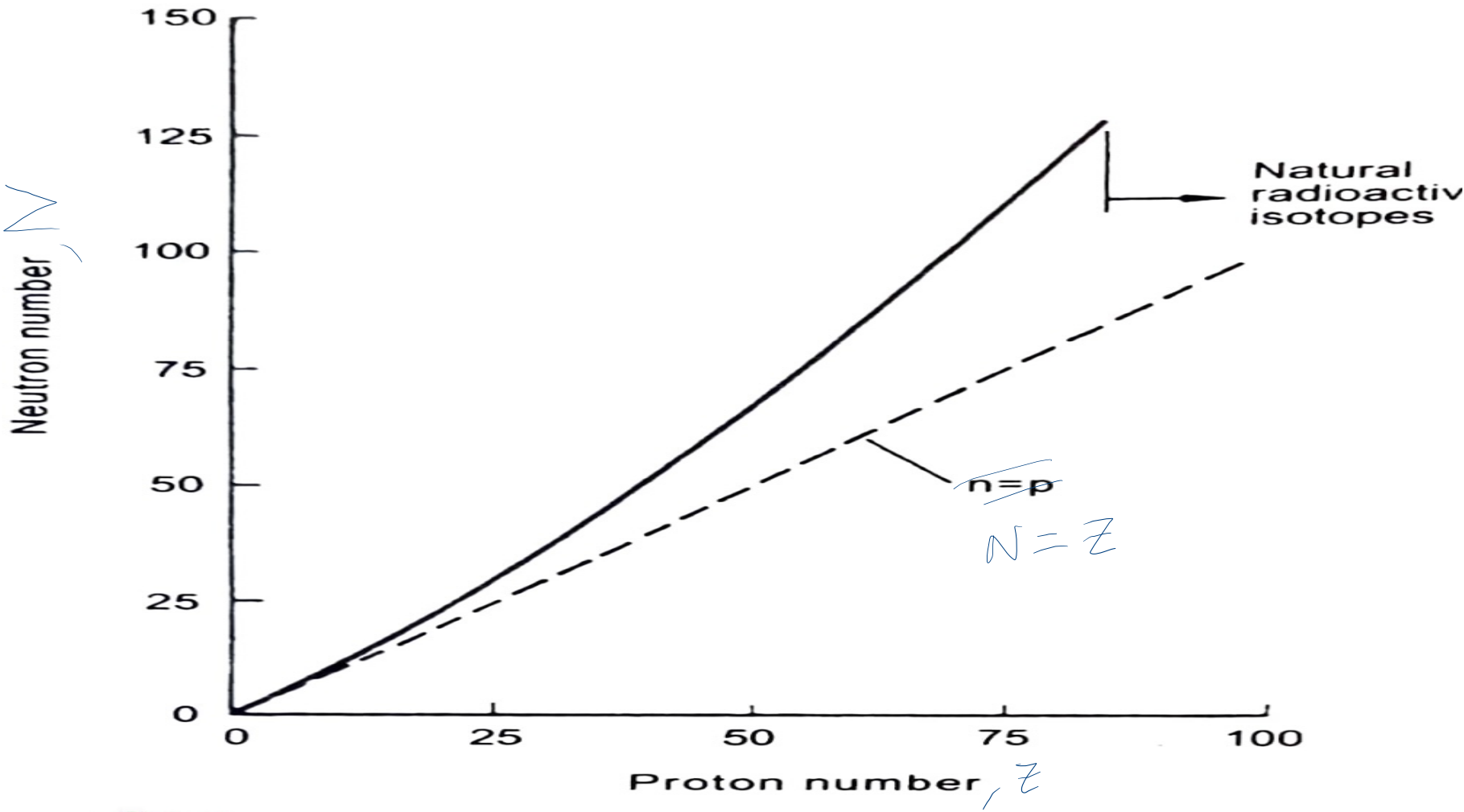
^{A}X

H

N

C

Nuclear Stability Curve



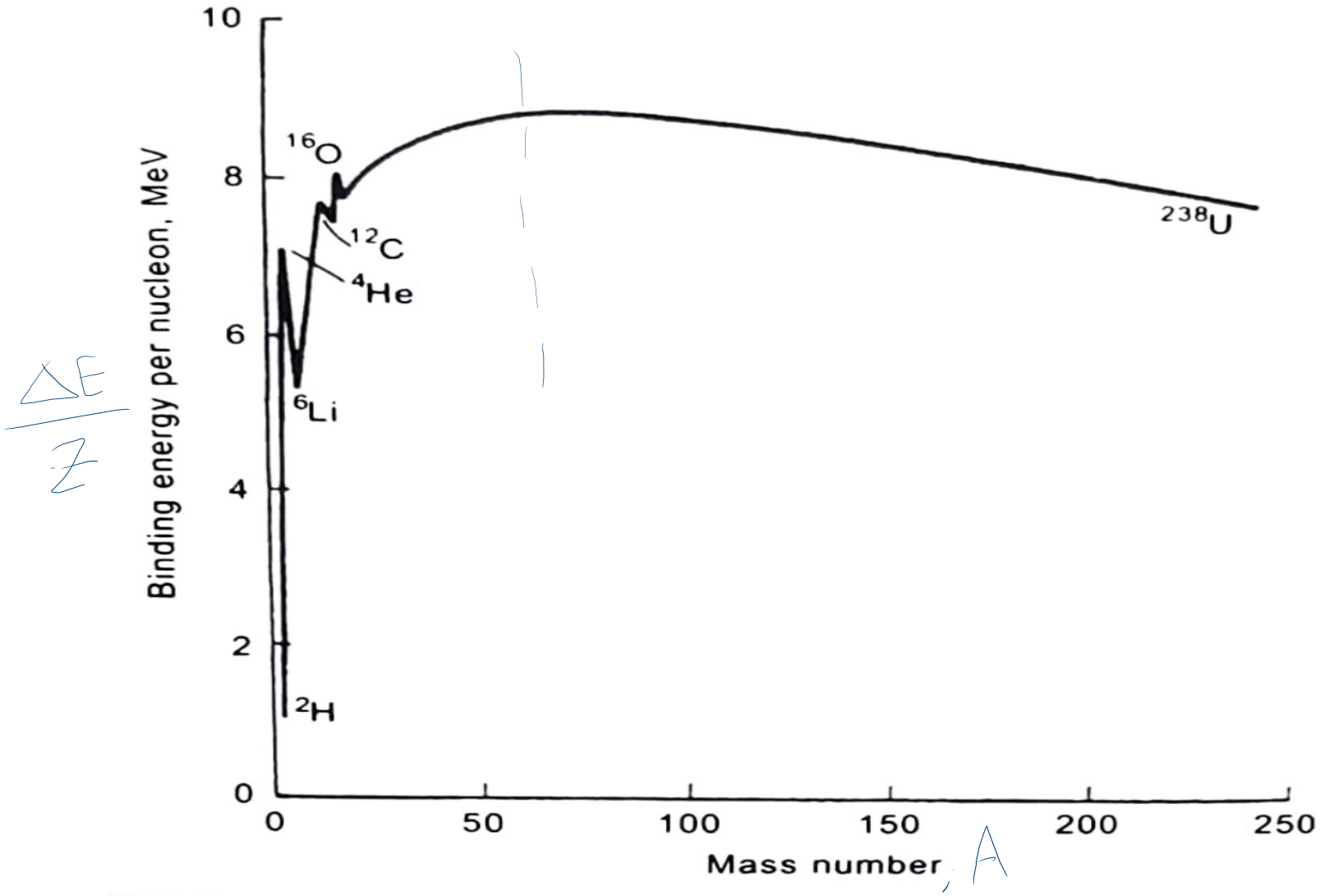
Binding Energy

1. At distances of the order of nuclear radius ($\approx 1.25 \times 10^{-15} A^{1/3}$ m) in a stable nucleus, Short range attractive nuclear forces are **stronger than** the Coulomb forces.
2. Hence, despite repulsive forces among protons, the protons and neutrons are held together in nucleus.
3. However, the ratio $\frac{A}{Z}$ increases with Z, i.e., the stable nuclides become increasingly **neutron–rich** with increasing Z.

Binding Energy

1. Mass defect $\Delta m = [Z\, m_P + (A - Z)\, m_N] - m_{nucleus}$
2. This mass defect is conceptually thought of as having been converted to energy, $E = \Delta m\, c^2$ at the time of formation of nucleus putting it in negative energy state.
3. The amount of externally supplied energy that would have to be converted to mass in disassembling a nucleus into its constituent nucleons is known as the **Binding Energy (BE)** of the nucleus.
4. $\mathrm{BE} = \Delta m\, c^2$

Binding Energy per Nucleon



Nuclear Reactions

- In study of nuclear reactors, the following two types of nuclear reactions are important:
  1. Spontaneous disintegration of nuclei, e.g., radioactive decay of fission products, and
  2. reactions resulting from collision between nuclei and/or nuclear particles, e.g., neutron induced nuclear fission
- **Spontaneous disintegration** depends purely upon the properties of an individual nucleus.
- **Nuclear particles–nuclei collision based interactions** depend upon
  - properties of the colliding particles/nuclei, and
  - relative velocities with which collision occurs.

Nuclear Reactions: 1. Radioactive Decay

- α decay

$$ {}^{A}_{Z}X \xrightarrow{\alpha} \underbrace{{}^{4}_{2}He}_{\alpha\text{-particle}} + {}^{A-4}_{Z-2}X $$

- β decay

$$ {}^{A}_{Z}X \xrightarrow{\beta} {}^{A}_{Z+1}Z + \beta^{-} $$

- γ decay

$$ {}^{A}_{Z}X \xrightarrow{\gamma} {}^{A}_{Z}X + \gamma $$

Nuclear Reactions: 1. Radioactive Decay

- $\frac{dN(t)}{dt} = \dot{N}(t) = -\lambda\, N(t)$ || Law of Radioactive decay

  $\lambda > 0$

- $N(t) = N(t_0)\, e^{-\lambda(t-t_0)}$

  $N(t) = N(0)\, e^{-\lambda t}$

  λ = decay constant s^{-1}

- Decay rate of nuclei $= -\lambda\, N(t_0)\, e^{-\lambda(t-t_0)}$

  $= -\lambda\, N(0)\, e^{-\lambda t}$

- Mean lifetime, $\bar{t} = \frac{1}{\lambda}$ s

- Half life, $T_{\frac{1}{2}} = \frac{ln\, 2}{\lambda}$

Nuclear Reactions: 2. Radioactive Decay + Production

- $\frac{dN(t)}{dt} = \dot{N}(t) = -\lambda\, N(t) + R(t)$

  decay rate production rate

- Radioactive decay chain: $X \longrightarrow Y \longrightarrow Z \longrightarrow$

$$
\begin{aligned}
\frac{d\,N_X}{dt} &= -\lambda_X\, N_X + R_X(t) \\
\frac{d\,N_Y}{dt} &= -\lambda_Y\, N_Y + R_Y(t) \\
\frac{d\,N_Z}{dt} &= -\lambda_Z\, N_Z + R_Z(t)
\end{aligned}
$$

$= \lambda_X N_X$

$= \lambda_Y N_Y$

Nuclear Reactions: 3. Nuclear Collision Reactions

target nucleus

incident particle

- The nuclear reaction $\quad$ a+b $\longrightarrow$ c+d

  is often written as $\quad$ a(b,c)d

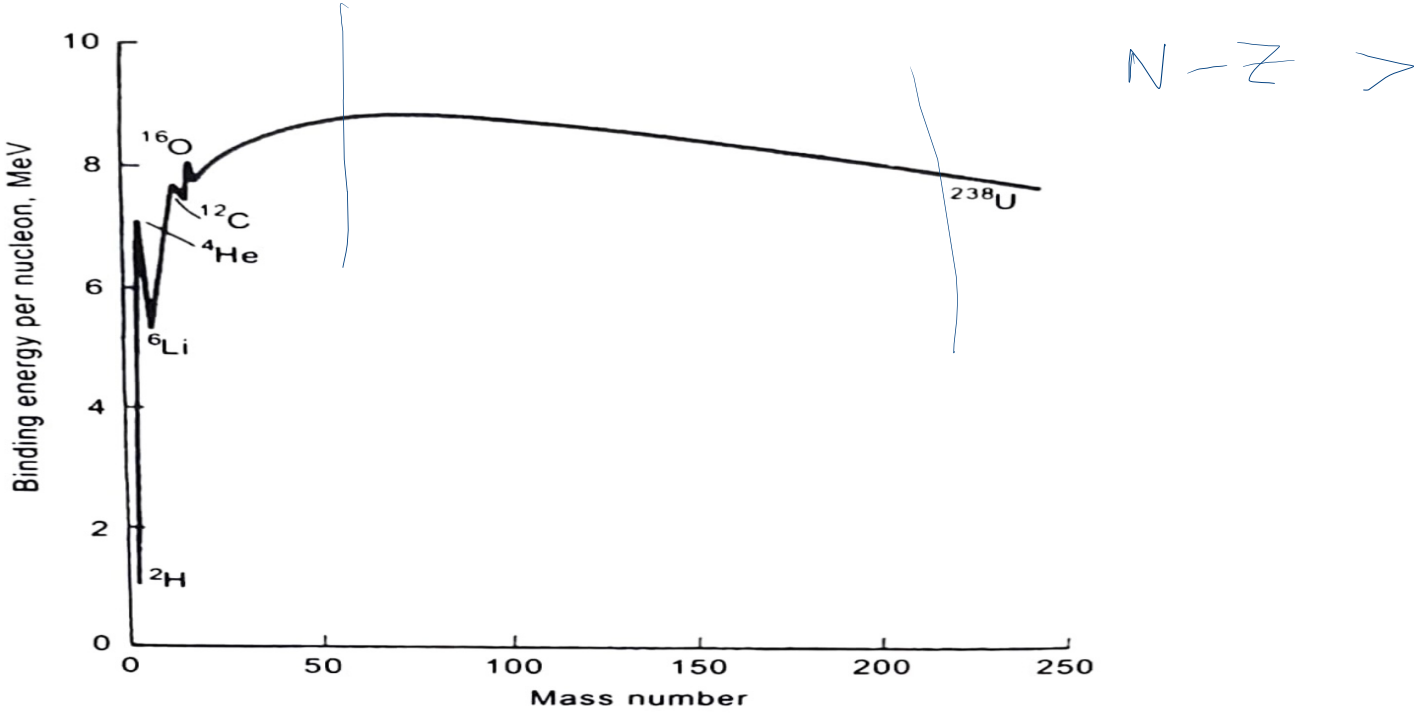
emitted particle/radiation

- Example: $\quad {}^{1}_{0}n + {}^{235}_{92}U \longrightarrow {}^{236}_{92}U + \gamma$

  is commonly written as $\quad {}^{235}_{92}U(n,\gamma){}^{236}_{92}U$

compound nucleus

Nuclear Reactions: 3.1 Neutron–Induced Fission



- When a neutron is bombarded on a target nucleus (A,Z), to form the compound nucleus (A+1,Z), then
- the BE per nucleon of the *compound* nucleus $<$ BE per nucleon of the *target* nucleus.

Nuclear Reactions: 3.1 Neutron–Induced Fission

- When a neutron is bombarded on a target nucleus (A,Z), to form the compound nucleus (A+1,Z), then
- the BE per nucleon of the *compound* nucleus $<$ BE per nucleon of the *target* nucleus.
- This reduction may be sufficient that the compound nucleus will undergo fission even if the incident neutron has low energy.
- Fissile nuclides (e.g., ${}^{233}_{92}U$, ${}^{235}_{92}U$, ${}^{239}_{94}Pu$, ${}^{241}_{94}Pu$) have high probability of undergoing fission on bombardment by low energy neutrons.
- Fissionable nuclides (e.g., ${}^{232}_{90}Th$, ${}^{238}_{92}U$, ${}^{240}_{94}Pu$) will undergo fission on bombardment by high energy neutrons

Nuclear Reactions: 3.3 Elastic Scattering

- The compound nucleus formation is followed by emission of a neutron that returns the compound nucleus to the ground state of the original nucleus.

$$ {}_0^1n + {}_Z^A X \longrightarrow \quad {}_0^1n + {}_Z^A X $$

$$ {}_Z^A X\,(n, n)\,{}_Z^A X $$

- The Kinetic Energy of the original neutron–nucleus system is preserved.

Nuclear Reactions: 3.3 Inelastic Scattering

- The compound nucleus formed by neutron capture decays by emission of one neutron, leaving the nucleus in excited state, which subsequently further decays.
- The energy of the emitted neutron can be considerably less than the energy of the incident neutron.

$$ {}^{1}_{0}n + {}^{A}_{Z}X \longrightarrow \quad {}^{1}_{0}n + \left({}^{A}_{Z}X\right)^{*} \qquad {}^{A}_{Z}X\,(n, n')\,{}^{A}_{Z}X^{*} $$

$$ {}^{1}_{0}n + {}^{A}_{Z}X \longrightarrow \quad {}^{1}_{0}n + {}^{A}_{Z}X + \gamma \qquad {}^{A}_{Z}X\,(n, n')\,{}^{A}_{Z}X $$

- The compoud nucleus may decay by emission of two of more neutrons. Such events are referred to as (n,2n), (n,3n), . . . events.

Nuclear Reactions: 3.4 Other Reactions

- (n, α) reactions

$$^{10}_{5}\mathrm{B} + ^{1}_{0}\mathrm{n} \longrightarrow ^{7}_{3}\mathrm{Li} + ^{4}_{2}\mathrm{He}$$

- (α, n) reactions

$$^{9}_{4}\mathrm{Be} + ^{4}_{2}\mathrm{He} \longrightarrow ^{12}_{6}\mathrm{C} + ^{1}_{0}\mathrm{n}$$

- (γ, n) reactions

$$^{2}_{1}\mathrm{H} + \gamma \longrightarrow ^{1}_{1}\mathrm{H} + ^{1}_{0}\mathrm{n}$$

$$^{9}_{4}\mathrm{Be} + \gamma \longrightarrow 2\ ^{4}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n}$$

- (γ, e) reactions

Compton
Photoelectric effect